Non-frontal face recognition method with a side-face-correction generative adversarial networks

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Abstract—**Frontal face image recognition is the main target of traditional face recognition.The deflection of the human face often causes the dislocation of the facial features,which leads to the reduction of the recognition accuracy of the non-frontal face.To solve the above problems,a non-frontal face recognition model based on generative adversarial network is proposed.In this model,the angle information is encoded separately by using a two-channel generator and auto-coding network,and the non-frontal face image in natural environment is corrected to obtain the frontal face image.Through the multi-discriminator mechanism of facial attention,we set discriminators in the eye, eyebrow, nose, mouth and the whole area of the face image so as to retain the details of the face to the maximum extent while ensuring the clarity of image.Then the corrected face features are extracted by Facenet and MTCNN to obtain the non-frontal face recognition results.The model is validated on multi-PIE dataset and CFP dataset.The results show that the accuracy of non-frontal face recognition is improved by 1% in CFP dataset compared with VGG-FACE, TP-CNN and HPN.**

Keywords-image processing; generative adversarial network; feature extraction; side face correction; recognition on non-frontal face;

I. INTRODUCTION

Face recognition plays an increasingly important role in today's society[1].Traditional face recognition methods are suitable for face images in experimental environment.However, in the natural environment,it is tough to maintain the front face all along.After the deflection of the facial posture,the facial organs are often displaced,and the self-shielding effect will also occur,which will lead to the loss of most features of the face. When the rotation angle of the face reaches 45° to 90°,the displacement of the facial organs becomes more significant and the accuracy of recognition is greatly reduced.

To solve the above problems,researchers proposed a 3D[2-4] modeling method,which can obtain a better standardized multi-pose face by building a 3D model of the face.However, 3D modeling based on a single 2D face image will generate too much noise information.Tal Hassner[5] believed that it would greatly increase the cost of the system to set up a 3D face shape for every input face images,so they conducted face detection along 49 facial feature points of the jaw bone,and then used a single unmodified 3D surface as the face shape of all input.Zhu[6] proposed a natural face automatic generation technology,which is a standardized algorithm using 3D deformation model (3DMM) to generate high-fidelity posture and facial expression.In order to enable the generator to synthesize positive face images,Hu[7] used the control signal to guide the generator,and proposed a CAPG-GAN model that can be rotated for faces at any angle based on this method.

Using conjunctive models[8-9] and specific attitude models[10-11],facial features can be directly extracted from non-frontal face images to achieve multi-angle face recognition.Face recognition technologies based on deep learning, such as VGGFace[12] and DeepFace[13],have achieved good results in multi-pose face recognition.However, both of them rely on large amounts of training data.In DeepFace,for example,the model's performance was significantly reduced when the face deflection angle varied between 45° and 90°.To solve the problem of low accuracy of face recognition under multi postures,Zhao[14] proposed a recognition algorithm based on dual path convolution neural network.This algorithm is based on a large number of face marker data and it has high requirements for GPU performance,the convergence speed of the model greatly depends on the parallel computing ability of GPU,and its recognition accuracy will be significantly reduced when the face deflection angle is greater than 60°.

In order to solve the above problems,we propose a non-frontal face recognition model based on generative adversarial network (GAN)[15].The work content and innovation of this paper are as follows:

We draw a lesson from generative adversarial networks(GAN),and we proposed a non-frontal face recognition method with a side-face-correction generative adversarial network(SFC-GAN).We use generative adversarial networks to correct non-frontal face images.After obtaining frontal face images,facial features can be extracted.Then we can contrast euclidean distance between extracted face features and features

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in the standardized database.Final result of recognition will be obtained.This method reduces the difficulty of correcting non-frontal face images.The accuracy of recognition can also be improved.

II. Materials and Methods

SFC-GAN consists of two parts:generation module and discrimination module.The traditional generator is changed to a self-coding structure for feature representation of the input images,the dual-channel mode is adopted to maintain the clarity of the generated frontal face images to the maximum extent.In the discrimination module,the traditional binary discriminator is changed into the attention multi-discriminant mechanism,and three discriminators for eyes,nose and mouth are added to the original whole region discriminator structure.The specific network structure is shown in Figure 1:

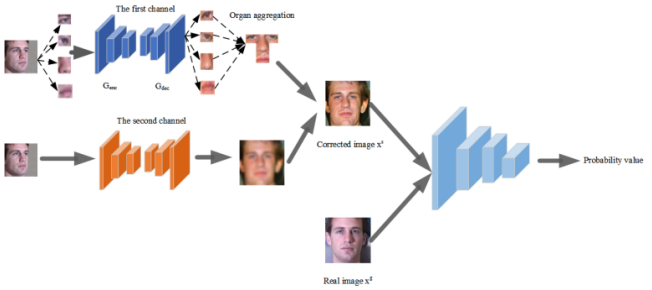


Figure 1. Network structure of side face correction model

A. Generation module

Non-frontal face recognition requires both contour and feature details.A pair of side face images were corrected in the first channel to obtain the frontal face contour.Channel 2 locates the five facial organs on the non-frontal face,cuts the facial organs,corrects them to frontal through the generation module,and then merges them with the facial contour generated by Channel 1.See Figure 2 for the network structure of the two-channel generator:

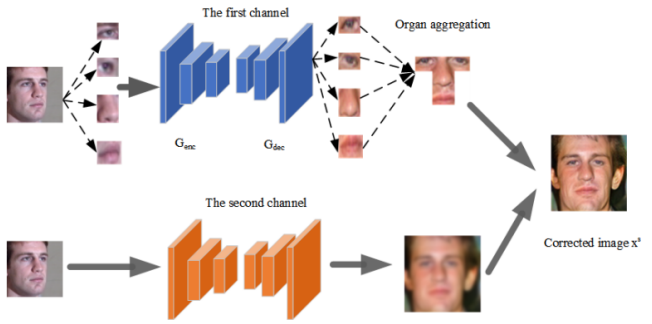


Figure 2. Network structure of dual-channel generator

LightCNN network framework is adopted as the face feature extraction framework.Encoder and decoder together constitute generators.Convolutional neural network is the core of the encoder, which can extract the corresponding features from the non-frontal face images.

The network structure is shown in Figure 3.In order to extract features from face images,the input is 128*128*3 non-frontal face images.Convolution layer and full connection layer constitute the network framework.The first half of the network is convolution layer,and the second half of the network is full connection layer.

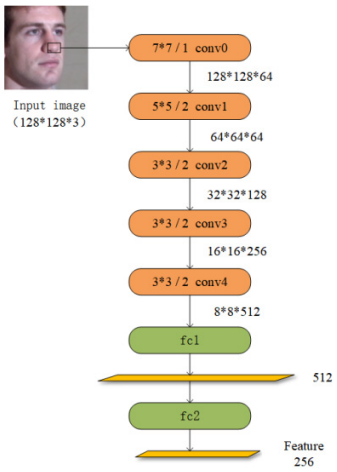


Figure 3. Network structure of encoder

In the decoder network,deconvolution neural network is its core.After obtaining facial features extracted by encoder,decoder can composite frontal face images.The network structure is shown in figure 4,and the decoding network is composed of full connection layer and deconvolution layer.

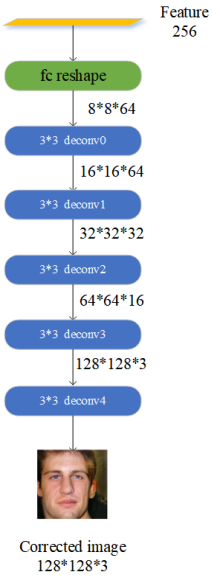


Figure 4. Network structure of decoder

B. Discrimination module

In order to form a game with the generation module and promote more realistic frontal face image generation,the discriminator needs to output a probability value.Through the probability value,we can distinguish whether the input of the discriminator is a natural face image or a face image synthesized by the generator.The larger the probability value is,the more likely the input image of the discriminator is not the real image,but the composite image generated by the generator.Our

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purpose is to obtain a smaller probability value when the input of the discriminator is generated image rather than real image.At this time,it can be proved that the face images synthesized by the generator are real enough,which confuses the judgment of the discriminator.The network structure of the discriminator is shown in Figure 5:

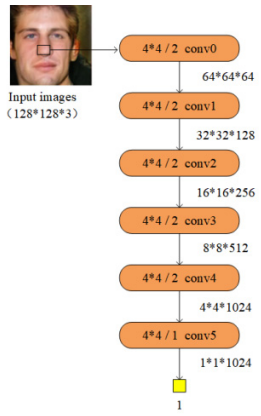


Figure 5. network structure of discriminator

In order to ensure that the facial features are preserved to the greatest extent in the process of side face correction,this model adds three local discriminators based on the original GAN network discriminator network structure,which are eyebrow discriminator,nose discriminator and mouth discriminator.In order to cut the specific location of the face images,the input of the global face discriminator is limited to 128*128.Due to the face image,each facial organ is in the same position with a high probability.In order to cut the facial organs, it is necessary to locate each part of the face image first.Since the size of facial organ is basically the same,the image size of eyebrows and eyes should be limited to 20*95,the image size of mouth should be limited to 26*40,and the image size of nose should be limited to 35*22.The structure of the discriminator is shown in Figure 6:

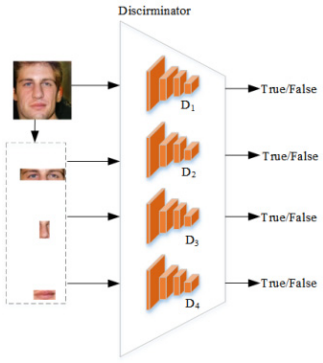


Figure 6. Network structure of discriminator

C. loss function

This model needs to correct non-frontal face images and generate frontal face images,so it needs symmetric loss function and pixel loss function.In the GAN network, it is necessary to optimize adversarial loss function.There are four discriminators,so the four adversarial loss functions are weighted summation.

Pixel loss function can be expressed as (1):

$$L_P = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} \left| G(x^p)_{i,j,k} - x^g_{i,j,k} \right| \quad (1)$$

The width of the image is W,and the height of the image is H.In order to represent the loss function, it is necessary to use x^p to represent the side face image of the experimental object,x^g to represent the real frontal face image,and $G(x^p)$ to represent the corrected face image generated by the generator.The difference between the real face image and the synthetic face image is the pixel loss function.The smaller the difference is,the closer the synthetic face image is to the real face image.Manhattan distance is used to represent the difference and the loss function L_p needs to be minimized.

Symmetry loss function can be expressed as (2):

$$L_s = \frac{1}{W/2 \times H} \sum_{i=1}^{W/2} \sum_{j=1}^{H} \left| G(x^p)_{i,j,k} - G(x^p)_{W-i,j,k} \right| \quad (2)$$

The symmetry loss function accelerates the convergence speed of the generator by calculating the Manhattan distance on the left and right sides of the synthesized face,retains the face feature to the maximum extent,and promotes the rapid synthesis of frontal faces.

Adversarial loss function of discriminator can be expressed as (3):

$$L_{c_k} = \ln D_k(x_k^g) + \ln(1 - D_k(G(x^p)_k)) \quad (3)$$

There are four discriminators in the model,so the sum of four adversarial loss functions needs to be weighted.In order to enable the discriminator to accurately determine the source of images,the discriminator training needs to maximize L_{c_k}.

Loss function of generator can be defined as(4).Loss function of discriminator can be defined as(5).

$$L_G = L_P + L_S + L_{C_k} \quad (4)$$

$$L_D = \varpi_1 L_{c_1} + \varpi_2 L_{c_2} + \varpi_3 L_{c_3} + \varpi_4 L_{c_4} \quad (5)$$

D. Face recognition

Face recognition mainly includes two parts:face detection and face recognition.The three-level convolutional neural network MTCNN can obtain the face marker points to obtain the face position,and then cut the face image to acquire face images with less noise.Facenet network accepts the clipped face images and extracts the feature vector of 128 length.

To form a standardized database,we will store the feature vectors of each experimental objects.The frontal face images in the dataset are input into the Facenet network.Then we can obtain face features and extract feature vectors.The non-frontal face images in dataset are input into SFC-GAN.After receiving corrected face images,we can use Facenet to extract features from these updated images.Euclidean distance between features extracted from corrected images and features in the standardized database will be calculated.

In order to form a standardized database,the feature vectors of each experimental object will be stored.The frontal face images in the dataset are input into the Facenet to obtain face

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features and extract feature vectors.The non-frontal face images in the dataset are input into the SFC-GAN network,and the non-frontal faces are corrected.After obtaining the corrected face images,the feature vectors are extracted by Facenet,and the Euclidean distance between the face features of corrected face images and the features in the standardized database is calculated.The structure of the face recognition network is shown in Figure 7.

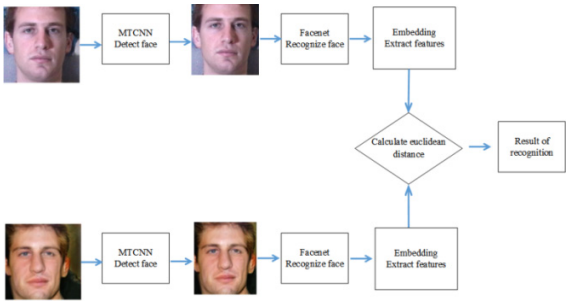


Figure 7. The process of face recognition

III. ANALYSIS OF EXPERIMENT

A. *Parameters for experiment*

Parameters required for the experiment are shown in table 1.

TABLE I. EXPERIMENTAL PARAMETER TABLE

Option	Configuration
Operating system	64bit Windows10
CPU	Intel(R) Core(TM) i7-9750H 2.60GHz
GPU	NVIDIA GeForce GTX 1650Ti
Framework	PyTorch
Environment	OpenCV 4.5.4、CUDA 11.2.162

B. *Database*

Multi-pie dataset contains 750,000 face images,which is a professional dataset for face recognition and expression recognition.Under the strict control of light conditions, researchers can obtain face images under different angles.There are 7000 images in the CFP dataset.CFP dataset includes frontal face images and non-frontal face images.There are 5000 frontal face images and 2000 non-frontal face images.Images in CFP dataset were collected in the natural environment.

C. *Evaluation of experimental results*

This experiment is divided into two stages.The non-frontal face images are corrected to obtain the frontal faces,and then the generated frontal face images are recognized through MTCNN and Facenet network.

1)Result of correction

Adam algorithm is used to optimize the network weight,and the learning rate is set to 0.0001.Model training is divided into two stages.In the early stage of training, the generator and the discriminator keep the same training frequency for training.In the later stage of training,the ability of discriminator and generator has been improved,mainly for the generator training,the number of generator training and discriminator training times is 2:1.

The face images of 200 volunteers in the Multi-pie dataset under seven different poses were used as the training set,and the face images of 49 volunteers under seven different poses were used as the test set.In order to reduce the influence of non-face region on the experimental process,it is necessary to preprocess the face and select the effective face region.All images are divided into standard face images of 128*128 size to reduce over-fitting during training.The experimental results are shown in Figure 8.In addition,a part of images in CFP were chosen to be tested.The experimental results are shown in Figure 9.

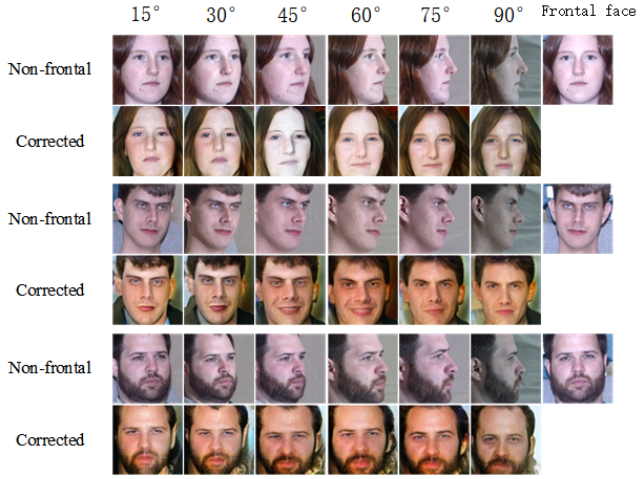


Figure 8. The correction results on Multi-pie dataset

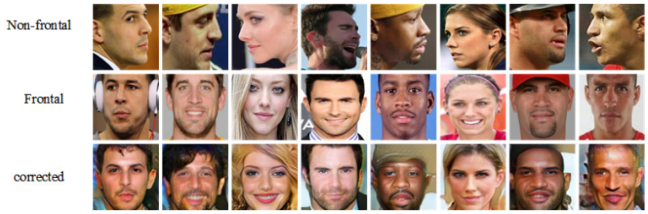


Figure 9. Correction results on CFP dataset

It can be seen that in the experimental environment and non-experimental environment,the corrected frontal face is clear and there is no large deformation.Compared with the frontal face image,the corrected images retain the effective face features for face recognition.When the face deflection angle is large,or even facial features cannot be easily extracted,the correction accuracy of the proposed method is still guaranteed.

2) Result of recognition

The experimental results should be accurate.The synthetic face images were divided into 10 groups,10 experiments were conducted,and the average value of 10 results was taken as the evaluation index.Other algorithms were selected for comparison to evaluate the effectiveness of the model,including HPN[16],VGG-FACE and TP-CNN,which did not use GAN

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network.FF-GAN[17] and MTAN[18] using GAN to correct non-frontal face images also included.Experimental comparisons on the Multi-pie dataset are shown in Table 2. Experimental comparisons on the CFP dataset are shown in Table 3.

TABLE II. RECOGNITION RESULTS ON MULTI-PIE DATASET

Method	Recognition rate/%						
	0°	*15°*	*30°*	*45°*	*60°*	*75°*	*90°*
VGG-Face	98.1	97.2	96.1	92.6	84.7	62.8	34.2
HPN	95.4	84.2	78.3	72.8	61.2	47.6	29.8
TP-CNN	97.1	96.7	94.4	92.9	87.8	75.6	59.8
FF-GAN	95.7	94.6	92.5	89.7	85.2	77.2	61.2
MTAN	96.5	95.3	89.7	87.9	84.1	-	-
Method	98.7	97.7	96.3	91.0	85.8	78.4	62.4

TABLE III. RECOGNITION RESULTS ON CFP DATASET

Method	Recognition rate/%	
	Frontal-Frontal	*Frontal-Profile*
VGG-Face	97.59	89.12
TP-CNN	97.35	92.41
HPN	95.13	86.57
Human	96.24	94.57
Method	95.74	93.50

It can be seen from Table 5 that in the Multi-pie dataset, compared with the FF-GAN method,when the face deflection is 90°,the recognition accuracy of the proposed method increases by 1.2%.It can be seen from Table 6 that on the CFP dataset,the proposed method improves the recognition accuracy of non-positive faces by 1%.It can be seen that the face recognition method proposed in this paper has certain advantages.When the input face pose angle increases,the method in this paper still has high recognition accuracy.

IV. CONCLUSIONS

We proposed a non-frontal face recognition method with a side-face-correction generative adversarial network.Combined with MTCNN and Facenet network,non-frontal face images will be corrected.After obtaining corrected images,we can get the result of recognition.The results show that,on the Multi-pie dataset and CFP dataset,accuracy of non-frontal face recognition can be improved efficiently.However,when the face has a non-horizontal deflection,our correction model cannot correct the face images.We will further enhance the non-frontal face recognition rate for face deflection at elevation angles.

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